higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ROADS
NQF LEVEL 3

11 MARCH 2019

This marking guideline consists of 5 pages.

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QUESTION 1

1.1
1.1.1 False
1.1.2 True
1.1.3 True
1.1.4 False
1.1.5 True
1.1.6 True
1.1.7 False
1.1.8 False
1.1.9 True
1.1.10 False

(10 × 1) (10)

1.2
1.2.1 Asphalt with aggregate ranges from 37,5 mm down to 0,07 mm or 26,5 mm down to 0,075 mm✓ and spaces between filled with fine aggregates✓

1.2.2 Showing direction of land marks and roads

1.2.3 Area from which some of the road material are taken

1.2.4 Slope of the road so that water runs from the centre of the road to the kerb or u-drain

1.2.5 Area between the road prism and boundary of road reserve

(5 × 2) (10)
[20]

QUESTION 2

2.1
- Valid certificate of compliance
- Free of leaks from fuel, oil or binders
- Straight and clean spray bar
- All spray heads the same, opening and closing at the same time, and not leaking when closed
- Under direct control of approved operator (5)

2.2
- Binder distributor
- Chip spreader
- Rotary broom
- Drag broom
- Pneumatic tire roller (Any 3 × 1) (3)

2.3 Nuclear device (1)

2.4 2.4.1 Concrete roads are more durable than asphalt roads.

2.4.2 Concrete roads have a longer life than asphalt road.

2.4.3 Concrete roads need less repair than asphalt roads.

2.4.4 Extreme weather conditions are liable to cause more damage to asphalt roads that to concrete roads.

2.4.5 Concrete roads are not damaged by oil leaks like asphalt roads.

(5 × 1) (5)

2.5 Edge restraint is the structure made of concrete placed at edge of the road. (1)

[15]

QUESTION 3

3.1
- Control sign✓ for example stop sign✓
- Command sign✓ for example turn left✓
- Prohibition sign✓ for example no bus✓
- Reservation sign✓ for example bus parking✓ (Any 2 × 2) (4)

3.2
- To give strangers information about the area.
- Remove distractions which will be caused when the driver tries to find his position.
- It also tells the name of places. (3)

3.3
- Shopping centres
- Loading area
- Junction/Crossing
- Next to school (Any 3 × 1) (3)

[10]

QUESTION 4

4.1 4.1.1 Combination of two or more simple curves in the same direction

4.1.2 Spiral curve consisting of simple and compound curve

4.1.3 Combination of simple curves in opposite direction

4.1.4 Arc with a constant radius between two straight lines of a road

4.1.5 Force necessary to keep an object moving in a curved path directed inward toward the centre of rotation

(5 × 2) (10)

4.2 Parabolic curve (1)

4.3 4.3.1 Point of vertical tangent
4.3.2 Point of vertical curve
4.3.3 Point of vertical intersection
4.3.4 Length of curve
4.3.5 Initial slope of roadway
4.3.6 Final slope of roadway
(6 × 1) (6)

4.4
- Poles
- Fish line (2)

4.5
- Optical micrometer
- Movable head
- Levelling screw
- Altitude bubble
- Telescope
- Eye piece
- Mirror (Any 6 × 1) (6)

[25]

QUESTION 5

5.1 5.1.1 Process of compressing the placed soil to increase soil strength and load capacity

5.1.2 Test used to calculate mechanical strength of road base courses and subgrades

5.1.3 Mass ratio of particular volume of a substance when compared with the canal volume of water

5.1.4 Test used on site to determine the dry density of soil or gravel

5.1.5 Device used to work out how dense a liquid is
(5 × 2) (10)

5.2 5.2.1 Gravel
5.2.2 Clay
(2 × 1) (2)

5.3
- To determine the maximum dry density achievable for the material being tested
- To determine the optimum moisture, content moisture, content of the soil sample (2)

5.4
- Modified AASHTO test
- Sand replacement density test
- California bearing ratio test
- Relative density test (4)

5.5 5.5.1 California bearing ratio
5.5.2 Relative density
(2 × 1) (2)

5.6
- Brush all loose material away using 100 mm paint brush.
- Place the ring of the density device on the surface.
- Place the dolly on the ring and partially drive the three pegs into the ground so that they just touch the ring preventing it from moving.
- Lightly tamp the dolly with the road tamper until the density ring is completely in contact with the ground.
- Drive in the nails until the ring is completely secured by the nails.
- Remove the dolly and brush the loose materials into the spoon or trowel.

(Any 5 × 1) (5)

5.7
- Loosen the in-situ material within the confines of the density ring.
- Remove the loosened material and the sand left in the ring and place it in the flat dish.
- Measure the average depth of the hole.
- Accurately weigh off 6 kg of density sand and place it on the pouring can.
- Place the density device on the density ring and fill the hole with the calibrated sand in the pouring can exactly as before. (5)

[30]

TOTAL: 100

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